Minqi's Solutions to the Exercises of Rick Durrett (2019), Probability: Theory and Examples (5th Edition)

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CHAPTER 1

Measure Theory

1.1. Exercise 1.1.1: An Example of Probability Space

Let $\Omega = \mathbb{R}, \mathcal{F} =$ all subsets so that A or A^c is countable, $P(A) = 0$ in the first case and $= 1$ in the second. Show that $(\Omega, \mathcal{F}, P)$ is a probability space.

Proof. First, we prove that $\mathcal{F}$ is a $\sigma-$algebra on $\mathbb{R}$.

(1) $\mathcal{F}$ is nonempty because, say, $\{\sqrt{2}\}$ is in $\mathcal{F}$.
(2) Suppose $A \in \mathcal{F}$, then, by definition of $\mathcal{F}$, $A = (A^c)^c$ or A^c is countable. Hence $A^c \in \mathcal{F}$.
(3) Suppose $A_1, A_2, \dots \in \mathcal{F}$ is a countable sequence of sets. Define

$$\begin{aligned}
\mathcal{A} &\equiv \{A_1, A_2, \dots\}, \\
\mathcal{B} &\equiv \{X \in \mathcal{A} : X \text{ is countable}\}, \\
\mathcal{C} &\equiv \mathcal{A} - \mathcal{B}.
\end{aligned}$$

If $\mathcal{C} = \varnothing$, then $\cup_{i=1}^{\infty} A_i$ is countable because of Theorem 2.12 of [**Rud76**]. Otherwise, pick any $A_m \in \mathcal{C}$, then A_m^c must be countable, because $A_m \in \mathcal{F}$ and $A_m \notin \mathcal{B}$. Since

$$\left(\bigcup_{i=1}^{\infty} A_i\right)^c = \bigcap_{i=1}^{\infty} A_i^c \subset A_m^c,$$

Theorem 2.8 of [**Rud76**] implies that $(\cup_{i=1}^{\infty} A_i)^c$ is countable. Therefore $\cup_{i=1}^{\infty} A_i \in \mathcal{F}$.

Next, we prove that P is a measure on $(\mathbb{R}, \mathcal{F})$.

(1) P is nonnegative because P maps to either 0 or 1. $P(\varnothing) = 0$ because $\varnothing$ is countable.
(2) Suppose $A_1, A_2, \dots \in \mathcal{F}$ is a countable sequence of disjoint sets. Define $\mathcal{A}, \mathcal{B}, \mathcal{C}$ as above. If $\mathcal{C} = \varnothing$, then, as we have shown above, $\cup_{i=1}^{\infty} A_i$ is countable. Hence

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = 0 = \sum_{A_i \in \mathcal{B}} P(A_i) = \sum_{i=1}^{\infty} P(A_i).$$

Otherwise, we claim that $\mathcal{C}$ cannot have two distinct elements. To show this, suppose $C_1, C_2 \in \mathcal{C}$ and $C_1 \neq C_2$. Then $C_1^c \cup C_2^c = \mathbb{R}$ because $C_1 \cap C_2 = \varnothing$. Then Theorem 2.8 of [**Rud76**] implies that $\mathbb{R}$ is countable, but this contradicts with the corollary of Theorem 2.43 of [**Rud76**].Therefore $\mathcal{C}$ contains only one element, say, A_m. Hence

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = 1 = P(A_m) = \sum_{A_i \in \mathcal{B}} P(A_i) + \sum_{A_i \in \mathcal{C}} P(A_i) = \sum_{i=1}^{\infty} P(A_i).$$

Finally, we prove that P is a probability measure. Note that $\mathbb{R}^c = \varnothing$ is countable. Hence

$$P(\mathbb{R}) = 1.$$

□

1.2. Exercise 1.1.4: The Union of Increasing $\sigma-$algebras is NOT a $\sigma-$algebra

(1) Show that if $\mathcal{F}_1 \subset \mathcal{F}_2 \subset \dots$ are $\sigma-$algebras, then $\cup_i \mathcal{F}_i$ is an algebra.
(2) Give an example to show that $\cup_i \mathcal{F}_i$ need not be a $\sigma-$algebra.

PROOF. (1) Pick any $F \in \cup_i \mathcal{F}_i$, then there exists i_0 such that $F \in \mathcal{F}_{i_0}$. Since $\mathcal{F}_{i_0}$ is an algebra, $F^c \in \mathcal{F}_{i_0}$. Hence $F^c \in \cup_i \mathcal{F}_i$.

Further more, pick any $E, F \in \cup_i \mathcal{F}_i$, then there exists i_0 and j_0 such that $F \in \mathcal{F}_{i_0}$ and $E \in \mathcal{F}_{j_0}$. Without loss of generality, suppose $i_0 \leqslant j_0$. Then $F \in \mathcal{F}_{j_0}$ because $\mathcal{F}_{i_0} \subset \mathcal{F}_{j_0}$. Since $F \in \mathcal{F}_{j_0}$ is an algebra, $E \cup F \in \mathcal{F}_{j_0}$. Therefore $E \cup F \in \cup_i \mathcal{F}_i$.

We have shown above that $\cup_i \mathcal{F}_i$ is closed under complements and pairwise unions, therefore it is an algebra.

(2) Let $\Omega \equiv [0, 1]$. Define

$$\begin{aligned}
\mathcal{F}_1 &\equiv \sigma\left([0.0, 0.1], [0.1, 1.0]\right), \\
\mathcal{F}_2 &\equiv \sigma\left([0.00, 0.01], [0.01, 0.10], [0.10, 0.11], [0.11, 1.00]\right), \\
&\dots.
\end{aligned}$$

where 0.1 is the binary representation of $\frac{1}{2}$, 0.01 is the binary representation of $\frac{1}{4}$, etc. Then it is clear that

$$\mathcal{F}_1 \subset \mathcal{F}_2 \subset \dots.$$

Pick an irrational number $x \in (0, 1)$, and suppose its binary representation is

$$0.s_1 s_2 s_3 \dots.$$

Then

$$[x, 1] = \bigcap_{i=1}^{\infty} [0.s_1 s_2 \dots s_{i-1} s_i, 1].$$

To arrive at a contradiction, suppose $\cup_i \mathcal{F}_i$ is a $\sigma-$algebra, then

$$[x, 1] \in \cup_i \mathcal{F}_i,$$

because $[0.s_1 s_2 \dots s_{i-1} s_i, 1] \in \cup_i \mathcal{F}_i$ for all i. Then there exists i_0 such that

$$[x, 1] \in \mathcal{F}_{i_0},$$

which contradicts with the obvious fact that $\mathcal{F}_{i_0}$ only contains intervals of rational endpoints.

□

1.3. Exercise 1.2.1: New Random Variable from Two Existing Random Variables

Suppose X and Y are random variables on (Ω, F, P) and let $A \in \mathcal{F}$. Show that if we let $Z(\omega) = X(\omega)$ for $\omega \in A$ and $Z(\omega) = Y(\omega)$ for $\omega \in A^c$, then Z is a random variable.

PROOF. Pick any $\omega \in \Omega$, then $\omega \in A$ or $\omega \in A^c$. Hence Z is defined on all $\omega \in \Omega$. Since both of X and Y map into $\mathbb{R}$, Z maps to $\mathbb{R}$. Therefore Z is a well-defined function from Ω to $\mathbb{R}$.

Next, we show that Z is a measurable map. Pick any $B \in \mathcal{R}$, then

$$\begin{aligned}
Z^{-1}(B) &= \{\omega \in \Omega : Z(\omega) \in B\} \\
&= \{\omega \in A : Z(\omega) \in B\} \cup \{\omega \in A^c : Z(\omega) \in B\} \\
&= \{\omega \in A : X(\omega) \in B\} \cup \{\omega \in A^c : Y(\omega) \in B\} \\
&= (A \cap \{\omega \in \Omega : X(\omega) \in B\}) \cup (A^c \cap \{\omega \in \Omega : Y(\omega) \in B\}) \\
&= (A \cap X^{-1}(B)) \cup (A^c \cap Y^{-1}(B)).
\end{aligned}$$

Since X and Y are random variables, $X^{-1}(B) \in \mathcal{F}$ and $Y^{-1}(B) \in \mathcal{F}$. Moreover, since $\mathcal{F}$ is an algebra, we conclude that

$$(A \cap X^{-1}(B)) \cup (A^c \cap Y^{-1}(B)) \in \mathcal{F}.$$

□

1.4. Exercise 1.2.2: Bounds of the Standard Normal Distribution Function

Let χ have the standard normal distribution. Use Theorem 1.2.6 to get upper and lower bounds on $P(\chi \geqslant 4)$.

Answer. Letting $x = 4$ in Theorem 1.2.6 of [**Dur19**] gives

$$(\frac{1}{4} - \frac{1}{4^3})e^{-8} \leqslant \int_4^\infty \exp(-\frac{y^2}{2})dy \leqslant \frac{1}{4}e^{-8}.$$

Since

$$P(\chi \geqslant 4) = \frac{1}{\sqrt{2\pi}} \int_4^\infty \exp(-\frac{y^2}{2})dy,$$

we conclude that

$$\frac{1}{\sqrt{2\pi}}(\frac{1}{4} - \frac{1}{4^3})e^{-8} \leqslant P(\chi \geqslant 4) \leqslant \frac{1}{\sqrt{2\pi}} \cdot \frac{1}{4} \cdot e^{-8}.$$

□

1.5. Exercise 1.2.3: A Distribution Function has at most Countable Discontinuities

Show that a distribution function has at most countably many discontinuities.

Proof. Pick any distribution function F, then, by Theorem 1.2.1 of [**Dur19**], F is monotonically increasing on $\mathbb{R}$. Theorem 4.30 of [**Rud76**] then implies that the set of points of $\mathbb{R}$ at which F is discontinuous is at most countable. □

1.6. Exercise 1.3.1: $\sigma(\mathcal{A}) = \mathcal{S}$ Implies $\sigma(X^{-1}(\mathcal{A})) = \sigma(X)$

Show that if $\mathcal{A}$ generates $\mathcal{S}$, then $X^{-1}(\mathcal{A}) = \{\{X \in A\} : A \in \mathcal{A}\}$ generates $\sigma(X) = \{\{X \in B\} : B \in \mathcal{S}\}$.

Proof. Note that,

$$\begin{gathered}
\sigma(\mathcal{A}) = \mathcal{S} \text{ and } \mathcal{A} \subset \sigma(\mathcal{A}) \\
\Downarrow \\
\mathcal{A} \subset \mathcal{S} \\
\Downarrow \\
X^{-1}(\mathcal{A}) \subset X^{-1}(\mathcal{S}) \\
\Downarrow \\
\sigma(X^{-1}(\mathcal{A})) \subset \sigma(X^{-1}(\mathcal{S})).
\end{gathered}$$

By the definition of $\sigma(X)$, $\sigma(X^{-1}(\mathcal{S})) = \sigma(X)$. Hence we conclude that

$$\sigma(X^{-1}(\mathcal{A})) \subset \sigma(X).$$

Next, we prove that $\sigma(X) \subset \sigma(X^{-1}(\mathcal{A}))$. Define

$$\mathcal{S}' \equiv \{B \in \mathcal{S} : X^{-1}(B) \in \sigma(X^{-1}(\mathcal{A}))\}.$$

Then,

$$\begin{gathered} X^{-1}(\mathcal{A}) \subset \sigma(X^{-1}(\mathcal{A})) \\ \Downarrow \\ \mathcal{A} \subset \mathcal{S}' \\ \Downarrow \\ \sigma(\mathcal{A}) \subset \sigma(\mathcal{S}') \\ \Downarrow \\ \mathcal{S} \subset \mathcal{S}'. \end{gathered}$$

Since $\mathcal{S}' \subset \mathcal{S}$, we conclude that $\mathcal{S} = \mathcal{S}'$. Hence, by the definition of $\sigma(X)$, we conclude that

$$\sigma(X) \subset \sigma(X^{-1}(\mathcal{A})).$$

□

1.7. Exercise 1.3.7: The Smallest Class of Simple Functions Closed under Limits

A function $\phi : \Omega \to \mathbb{R}$ is said to be simple if

$$\phi(\omega) = \sum_{m=1}^{n} c_m 1_{A_m}(\omega)$$

where c_m are real numbers and $A_m \in \mathcal{F}$. Show that the class of $\mathcal{F}$ measurable functions is the smallest class containing the simple functions and closed under pointwise limits.

Proof. Since $\phi(\omega)$ is $\mathcal{F}$ measurable if and only if the sets $A_1, \ldots, A_n$ are $\mathcal{F}$ measurable, it is clear that the class of $\mathcal{F}$ measurable functions contains all simple functions.

Pick any $\mathcal{F}$ measurable function X. Then Theorem 11.20 of [**Rud76**] implies the existence of a sequence $\{s_n\}$ of simple functions such that

$$\lim_{n\to\infty} s_n(\omega) = f(\omega), \quad \forall \omega \in \Omega.$$

Hence X is a member of any class containing the simple functions and closed under pointwise limits. Therefore the class of $\mathcal{F}$ measurable functions is a subset of any class containing the simple functions and closed under pointwise limits. □

Bibliography

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